

ABSTRACT

ProComp introduces a problem-driven multi-agent review paradigm that replaces unstable absolute scoring with structured analysis and comparative ranking. By integrating significance assessment, methodological validation, and contribution synthesis, it delivers more coherent and interpretable evaluations

PROBLEM STATEMENT

Peer review is overwhelmed, with 200,000+ annual submissions, long delays, and high reviewer disagreement. Existing LLM approaches remain unreliable, with direct prompting at MAE 1.844 (~30% accuracy) and prior multi-agent systems only modestly improving to MAE 1.464. ProComp introduces a problem-driven, sequential agent pipeline that achieves state-of-the-art MAE 1.396, enabling scalable, expert-aligned automated review.

METHODOLOGY

- **Research Paper + RAG** context are first read to ground the review in relevant background.
- Three specialists Agents evaluate the work from different angles: **The Problem, The Method, and The Contribution** and produces structured notes capturing strengths, weaknesses, and dimension-specific scores.
- **A Meta-Reviewer** combines all three into one coherent overall evaluation with a final score and justification

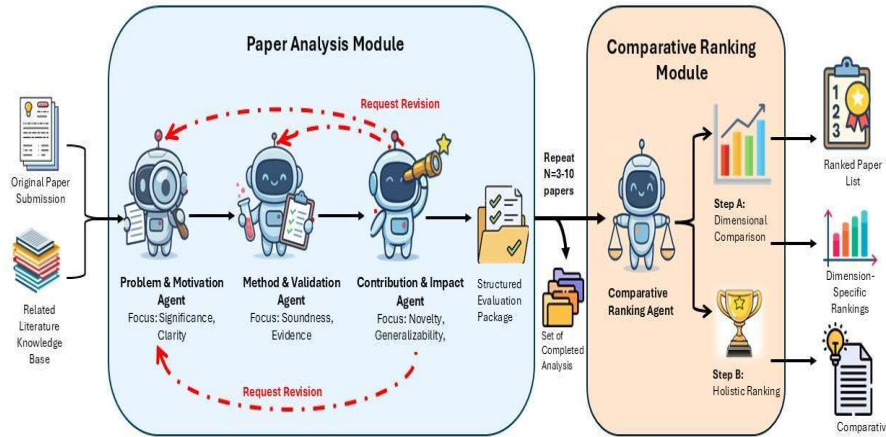


Figure 1: End-to-end ProComp workflow illustrating paper ingestion, specialized agent analyses, meta-review synthesis, and final score prediction

CONCLUSION

ProComp shows that structured, multi-agent evaluation reliably separates weak from strong work by scoring each scholarly dimension independently and producing clear, human-aligned quality distinctions. This consistent, dimension-specific sensitivity demonstrates that specialization provides a more rigorous and interpretable foundation for automated peer-review than undifferentiated single-prompt scoring.

REFERENCES

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- Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)* (pp. 1–22). Association for Computing Machinery. <https://doi.org/10.1145/3586183.3606763>

LIST OF RESEARCH QUESTIONS

- Q1 : How effectively can specialized agents capture key aspects of a paper, such as problem significance, methodology rigor, novelty, and results impact?
 Q2 : How reliably do LLM review scores preserve the ordinal structure of paper quality groups through discriminative scoring behavior?
 Q3 : How well does the multi-agent evaluation system is better while compared with direct prompting and existing baseline methods?

RESEARCH QUESTION 1

ProComp's multi-agents capture key aspects by isolating each evaluation dimension and scoring it independently.

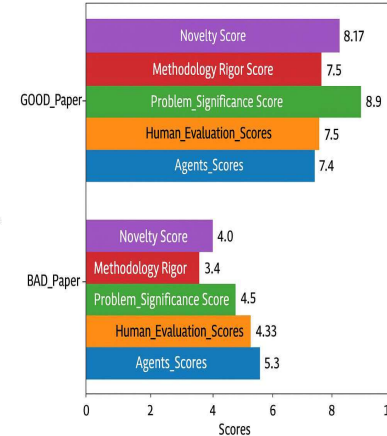


Figure 2: Comparison of GOOD and BAD paper scores across five evaluation dimensions, showing consistent quality separation

RESEARCH QUESTION 2

LLM scores tightly cluster within each quality group and cleanly separate across groups, and the box plot makes this explicit by showing low variance for BAD/BORDERLINE/GOOD clusters and clear ordinal spacing between them.

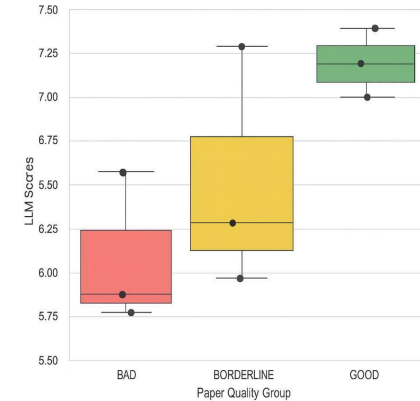


Figure 3: LLM score distributions for BAD, BORDERLINE, and GOOD papers, showing clear discriminative ability through consistent ordinal separation.

RESEARCH QUESTION 3

ProComp's multi-agent system achieves state-of-the-art accuracy (MAE 1.396, 50%), outperforms six baselines – including deep_reviewer (MAE 1.564) and zero-shot prompting (MAE 1.675) with 10.7% error reduction

Table 1: Proposed Problem-driven Comparative (ProComp) Review Agents vs State-of-the-Art Baselines (N=30 stratified papers)

Baseline Methods	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	Accuracy
Agent Review	1.542	1.942	0.467
Cycle_reviewer	1.639	2.016	0.400
Chain.of.thought	1.844	2.104	0.300
Deep_reviewer	1.564	1.851	0.500
Tree Review	1.535	1.876	0.433
Zero_shot	1.675	2.009	0.367
ProComp	1.396	1.754	0.500